Quantum Coin Flip

Day 19 | 30-Day Quantum Computing Challenge — Mini Project

Quick Recap. A classical coin has two possible outcomes. A qubit can be prepared in a superposition; when measured, it produces one classical outcome according to its quantum-state probabilities.

1. Objective

Build a one-qubit circuit, apply a Hadamard (H) gate, perform 1,000 measurements, and compare the observed frequencies of 0 and 1 against the theoretical 50% each.

2. Circuit and Explanation

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|0> ---- [ H ] ---- [ Measure ] ---- c
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The qubit begins in |0>. The H gate creates an equal superposition of |0> and |1>. Measurement returns either 0 or 1. Repeating the preparation and measurement 1,000 times produces a distribution.

3. Essential Mathematics

Initial state: $|0\rangle = [1, 0]^T$

Hadamard: H = (1/sqrt(2)) [[1, 1], [1, -1]]

After H: |psi> = (|0> + |1>) / sqrt(2)

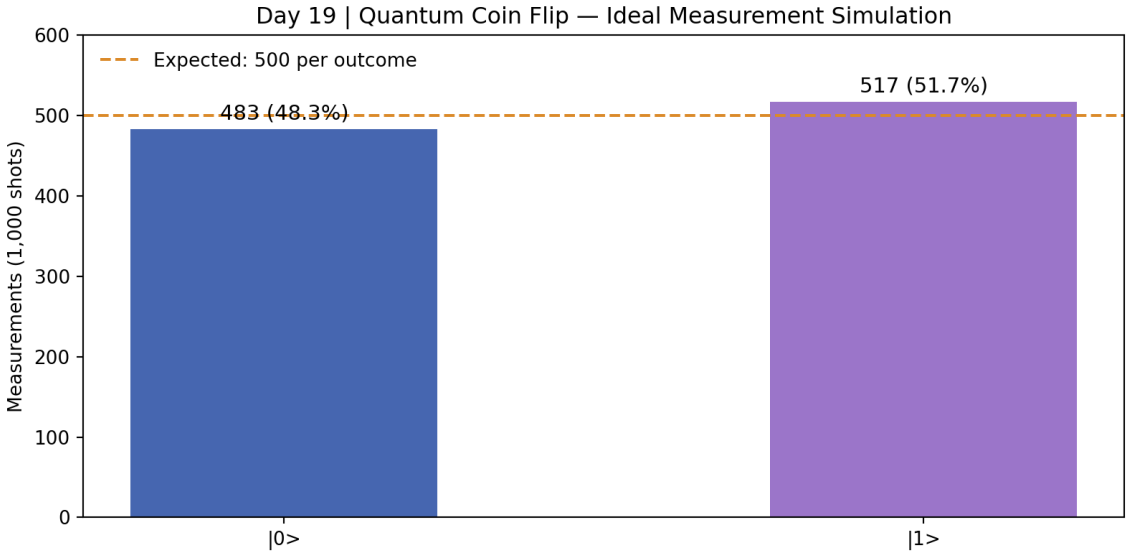
Born rule: $P(0) = |1/\sqrt{2}|^2 = 0.5$; P(1) = 0.5.

4. Execution Steps

1. Open **quantum_coin_flip.ipynb** in Jupyter or Colab.
2. Run the NumPy statevector and ideal measurement cells.
3. Inspect the counts and percentage distribution.
4. View or save **results.png**.
5. Optionally install Qiskit and Qiskit Aer to run the corresponding quantum circuit.

5. Sample Results

The sample seeded ideal measurement simulation produced **483 zeros (48.3%)** and **517 ones (51.7%)** from 1,000 shots. These counts may change with the random seed or execution backend.



6. What Does This Demonstrate?

The expected 50/50 split comes from the quantum state amplitudes. A finite number of measurements usually gives approximate, not exact, agreement. This is an ideal simulation; it does not establish the quality of a physical random-number source.

7. Common Mistakes

• Expecting exactly 500 of each outcome every time.
• Treating the Hadamard gate itself as a measurement.
• Calling a pseudorandom ideal simulator a certified hardware QRNG.
• Forgetting to repeat state preparation for each shot.

Summary: What to Remember

Start in |0> → apply H → obtain equal theoretical probabilities → measure repeatedly → compare the sample frequencies. **Next: Day 20 explores how the frequencies converge as shot count increases.**